CPE134 [Fundamentals of Electrical Circuits]

SUPERPOSITION THEOREM

LABORATORY EXERCISE 4

Leader: _____

Date: _____ Rating: _____

Members: _____

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NEW MESSAGE

From: **Neha Gupta** (Test Engineering Manager)

Subject: **Applying Superposition Theorem**

Good day, _____

I wanted to extend my appreciation for your excellent work on the testing procedure for our latest product, "Lotus". Implementing the KCL/KVL methodology has been a game-changer for our test engineering team, saving us a significant amount of time.

I'm reaching out once more for your expertise. Our customer wants to add the **superposition** functionality test for the product code: "Lily".

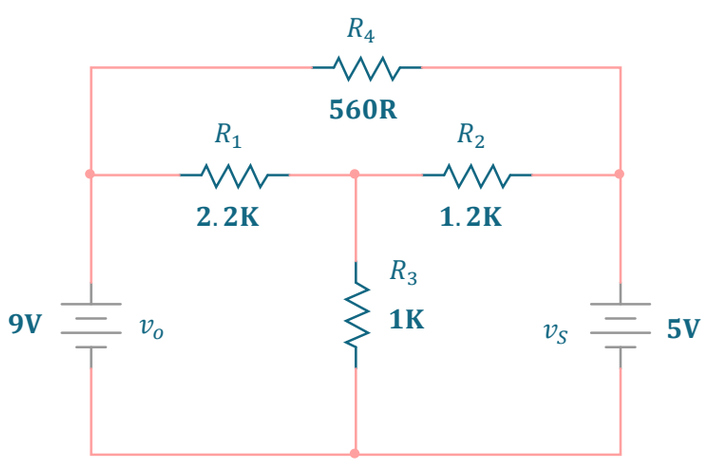
Please take a moment to review the attached files for more detailed information regarding this request.

Thank you for your ongoing support.

Best regards,
Neha

EXERCISE 1 VOLTAGE AND CURRENT MEASUREMENTS

Step 1 **Label** the variables and **set-up** the given circuit.



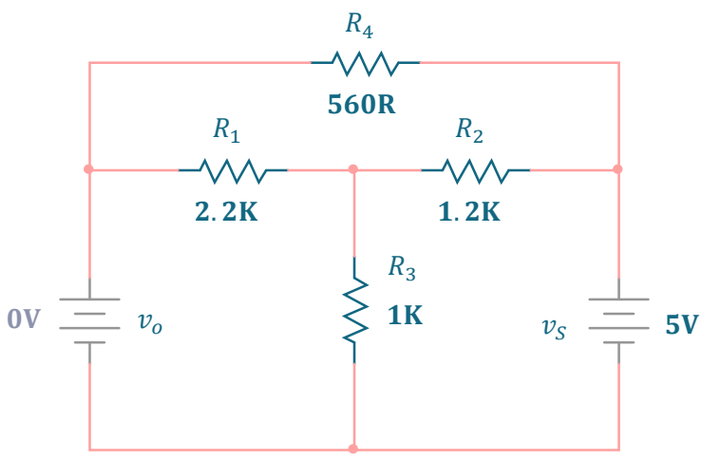
Step 2 Measure and record the actual **resistance**,**voltage** and **current** values.

Table 1

Parameters	Value
R_1	
R_2	
R_3	
R_4	
v_1	
v_2	
v_3	
v_4	
i_1	
i_2	
i_3	
i_4	

EXERCISE 2 DEMONSTRATING SUPERPOSITION THEOREM

Step 1 **Set $v_o = 0$**

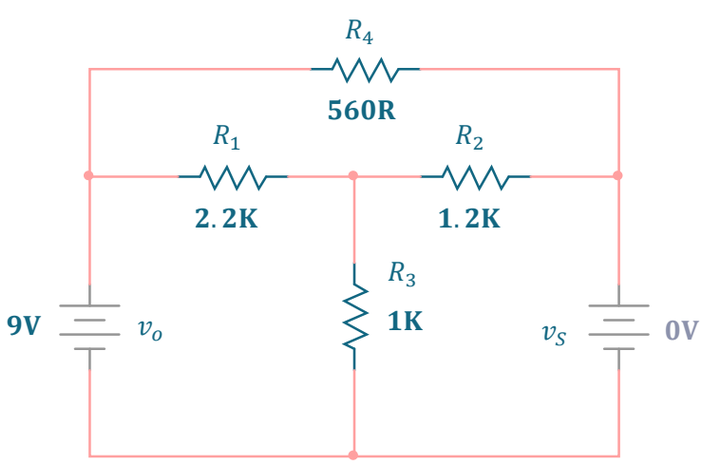


Step 2 Measure and record the actual **voltage** and **current** values.

Table 2

Parameters	Value
v_1'	
v_2'	
v_3'	
v_4'	
$i_{1'}$	
i_2'	
i_3'	
i_4'	

Step 3 **Set $v_S = 0$**



Step 4 Measure and record the actual **voltage** and **current** values.

Table 3

Parameters	Value
$v_{1''}$	
$v_{2''}$	
$v_{3''}$	
$v_{4''}$	
$i_{1''}$	
$i_{2''}$	
$i_{3''}$	
$i_{4''}$	

Step 5 Calculate the difference between Table 2 and Table 3 values.

Table 4

Parameters	Table 2	Table 3	Difference
v_1			
v_2			
v_3			
v_4			
i_1			
i_2			
i_3			
i_4			

Conclusion,